Fractal Dimension: Theory and Computation

Box-Counting Estimation via Hilbert Curve Analysis

ME

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1 Introduction

This is a test.

2 Measure Theory

The material in this section is standard and follows the presentation in [Men26].

First to define measure theory (I didn't take Lebesgue so I will be using the above notes...)

A measure is a way to assign a "size" to subsets of a set X. Let $\mathcal{F}$ be a σ-algebra on X.

Definition 2.1. (Measure). Let $(X, \mathcal{F})$ be a measurable space. A measure on (X, $\mathcal{F}$) is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

1. $\mu(\emptyset) = 0$
2. **Countable additivity:** if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

The triple $(X, \mathcal{F}, \mu)$ is called a **measure space**.

Proposition 2.2 (Basic properties)**.** *Let $(X, \mathcal{F}, \mu)$ be a measure space.*

1. ***Monotonicity:***

$$A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$$

2. ***Countable subadditivity:***

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

3. ***Continuity from below:*** *if $A_1 \subseteq A_2 \subseteq \cdots$ then*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{i \to \infty} \mu(A_i)$$

4. ***Continuity from above:*** *if $A_1 \supseteq A_2 \supseteq \cdots$ and $\mu(A_1) < \infty$ then*

$$\mu\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{i \to \infty} \mu(A_i)$$

The canonical example is the **Lebesgue measure** λ^n on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, where $\mathcal{B}(\mathbb{R}^n)$ is the Borel σ-algebra — the smallest σ-algebra containing all open sets. Lebesgue measure assigns to each box $[a_1, b_1] \times \cdots \times [a_n, b_n]$ its volume $(b_1 - a_1) \cdots (b_n - a_n)$, and extends uniquely to all Borel sets. It formalises the intuitive notions of length, area, and volume.

The inadequacy of Lebesgue measure for fractal geometry is immediate: a curve $\gamma \subset \mathbb{R}^2$ has $\lambda^2(\gamma) = 0$ (zero area) but $\lambda^1(\gamma)$ may be infinite (infinite length). Neither gives useful information about the "size" of γ. What is needed is a family of measures parametrised by a continuous exponent $s \geq 0$.

To construct such a family, we need the notion of an outer measure, which is slightly weaker than a measure — it is defined on *all* subsets of X, not just measurable ones, and only requires subadditivity rather than full additivity.

Definition 2.3 (Outer measure)**.** A function $\mu^* : 2^X \to [0, \infty]$ is an **outer measure** on X if:

1. $\mu^*(\emptyset) = 0$
2. **Monotonicity**: $A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B)$
3. **Countable subadditivity**:

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$$

The relationship between outer measures and measures is given by the Hahn-Carathéodory extension theorem: given an outer measure μ^* on X, the collection

$$\Sigma = \{A \subseteq X : \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A) \text{ for all } B \subseteq X\}$$

is a σ-algebra, and $\mu^*|_\Sigma$ is a measure. Sets in Σ are called **μ^*-measurable**. In particular, all Borel sets are μ^*-measurable whenever μ^* is constructed from a Borel-compatible pre-measure — which will be the case for Hausdorff measure.

The standard way to construct an outer measure is via coverings. If $\mathcal{K} \subseteq 2^X$ covers X (with $\emptyset \in \mathcal{K}$) and $\tilde{\mu} : \mathcal{K} \to [0, \infty]$ assigns sizes to sets in $\mathcal{K}$ with $\tilde{\mu}(\emptyset) = 0$, then

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \tilde{\mu}(K_i) \;\middle|\; K_i \in \mathcal{K},\ A \subseteq \bigcup_{i=1}^{\infty} K_i \right\}$$

defines an outer measure on X. This is precisely the construction used to define Hausdorff measure — with $\mathcal{K}$ being all subsets of diameter at most δ, and $\tilde{\mu}(U) = |U|^s$.

3 Hausdorff Measure

References

[Men26] Angeliki Menegaki. *Measure Theory and Integration: Lecture Notes MATH50006*. 2026.